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说话人 1 00:04:12

Morning. This team is. Everybody can hear me.Yep, we can hear you.Okay, perfect.So I just want to check for today's meeting, is Andrew and Steve or Mark will be on here? Oh, I saw Andrew here. Okay.

说话人 1 00:04:59

Shall we get started?

说话人 1 00:05:09

We may, yeah, we can go ahead and get started.Okay, so yeah, I just want to give a like highlight overview of like for today's meeting. And I know Lily's Lily sent us like.List of the getting items like for spp as pre P P Q and also for purification demo.So we know like those are pretty high priority.So we want to maybe tackle those first for today's meeting. So we prepared couple topics. We want to.Within Lily's opinions for today's meeting, the first is the cleaning program.And the second is the water system, and third is the welding and also the pass vision status and timeline.And also we want to talk about the glycol use and removal.So those couple topics want to just like see if Lily's team.Are okay with today's topic.

说话人 2 00:06:29

Yeah, sounds good.I think that's, yeah, so good plan, especially given those are fairly rate limiting here to respective starts.So sounds good.Okay, perfect.So yeah, we'll have a Doctor Joe to talk about the cleaning program first.

说话人 2 00:06:43

Okay, for the cleaning, you know, according to the discussion last in a previous meeting, we shared.The planned cleaning SP and the planned cleaning record to Ella Lily.So first I want to know, is this template or this at the P record looks good or if there is any comment from Ella Lily. So.If after we finalized on this SPP's, the cleaning sop and record, we can proceed with the, you

know, the following climate purification and, you know, the participation and join cleaning record and SOP. So I don't know whether you have got a chance to have a look at the eh provided. Cleaning SCP and Blackbird.

说话人 2 00:07:52

So, no, I haven't had a chance yet to review this. Okay, so from speaking, you know, we finished the cleaning SOP and clean record for the SPPS step. So are you are you sharing your screen? No, we're not share the screen because I'm sorry, I apologize. Yeah, because in the document, we have already shared with it in the Lily on the Sharepoint. So after the, sorry, you're talking about that you're sorry, you're talking about the document suite shared as part of the Spbs clean invalidation. Yeah.

说话人 3 00:08:33

Yeah, I know we've reviewed sections of those.

说话人 2 00:08:35

If we're just sections lose. Yeah, for the, yeah, thanks for your review of the document. I want to know if there is any comment to the cleaning sop and cleaning record for the SPP's equipment. If that's fine, we will proceed to prepare it.

说话人 2 00:09:04

Cleaning SOP and cleaning record. For the other steps, for example, the creativity equipment cleaning and the prepare purification, the participation and you know, the join equipment. Yeah, so I know we provided specific.

说话人 3 00:09:25

Recommendations and feedback on your cleaning procedures that pertain to the spp. Yes, sections before in our initial reviews. So if you're going to go about compiling, cleaning, validation documentation for the downstream activities, if you just read back to that initial feedback, I don't know a lot of it. I can't remember a lot of it. I'll talk my head. But there was sections there about your grouping methodology and some of the cleaning calculations. That you said as use previously, some of them were acceptable to some of them weren't. So if you just reference back to the feedback provided for the spp sections in your compiling of the downstream cleaning validation documentation, I think that would be a good start.

说话人 2 00:10:12

Yeah, it looks like you have already provided the comment to the, you know, to the clean SOP and record. So we will look at the share point and then we will improve accordingly and then have another share with you after our improvement.

说话人 4 00:10:35

Yeah, so first for step 1 for SPPS, so the cleaning validation protocol is Lily site for final confirmation. So we haven't fully confirmed it yet. And then yesterday, you, Wang, sent us on the storyboard for equipment cleaning. So that's currently in with Lily review at the moment. So in terms of confirming the SPPS Step 1 CVP. Will be next week. At some stage, we just need to double check our final comments. But a lot of the earlier comments have been added into it from the first review anyways from Lily. Okay, thank you for your review on the clean writing protocol. But now we're talking about, you know.

说话人 2 00:11:15

The cleaning SOP for SPPS equipment and cleaning record for SPPS equipment. You know, in a previous meeting, we talk about, you know, the equipment cleaning SOP is not in detail and the record is not in detail from the cleaning record and cleaning SOP, we cannot trace. Each small component of on equipment. So after that, we improved our cleaning SOP and cleaning record to ensure the cleaning SOP is in detail and the record can capture all the cleaning activities. So now I'm talk about the cleaning sop and cleaning record. Not in Canadian validation protocol. Okay, no problem. Thanks. Okay, sorry.

说话人 2 00:12:13

So maybe she, you know, you can look at the Sharepoint, whether the document are successfully uploaded to the Sharepoint. And if that's near, you can, you know, you can. You can remind, you know, either Isom Camp or you let it so we can have a review on the proposed cleaning sop and cleaning record. After that, we will work on the cleaning sop and clean record for the other steps. Okay, okay. So maybe she after wire, yeah, you can pull out the clean sop from your computer and share on the screen so everybody we can have a quick look.

说话人 2 00:13:28

Okay, yeah, so here you can see this is the, okay, we can share the screen for a moment. So here you can see this, the proposed cleaning equipment, but cleaning record, I second, sorry.

说话人 1 00:13:48

Share the wrong screen.

说话人 2 00:13:59

So, okay, okay. So here you can see this is the document we uploaded to the Sharepoint and that's about the equipment. But cleaner record, for example here, that's for the Spbs step. So they Inc here included all the. Cleaning procedure for each specific equipment include the equipment itself and the collection pipeline, the valve and or the consumables. For example, the filter and the, you know, only other component, not only include the cleaning, but also include the detection or operation.

说话人 2 00:14:43

The earning product. So here you can see at eh, eh in the right corner, that's the cleaner record. So in the middle, that's cleaning procedure. So in the right corner, that's the cleaner record. That's means when we do the cleaning, there's a cleaner method. And he on the right, there's a cleaner record. So this document, according to their method. So this is the our design of the cleaning method and clean record. But anyway, after your review, you can provide your comment so we can improve.

说话人 3 00:15:26

Even there is any comment. Yeah, like we can take this away and have a look at it, Alex. Okay. Okay. Thank you.

说话人 2 00:15:46

So next topic, let's the pure water, the host for the pure water. You know, in the previous meeting, we talk about how to plan the host or piping for the pure water between the use point and the distribution point. Here we propose the three approach.

说话人 2 00:16:09

So here you can see in the first table, you can see there is the three approach. The approach one, we use the current host collection between the distribution point and the use point. And then the approach 2, let's the silicon host with t wave. And approach three, we will introduce the fixed hard stainless steel pipeline between the use point and the distribution point.

说话人 2 00:16:41

So in each approach, what's the potential risk in place and what's the mitigation plan at ACM Camp? And what's the risk after the mitigation plan, and what's the due date of each of the approach.

说话人 2 00:17:00

So here you can see, for example, in the mitigation plan for the approach 1, there's not only this included the qualification of the worst case scenario of the sample point. From the end of the collected multiple host.

说话人 2 00:17:18

And we were so improved our procedure to define the requirement of a drain flush clean and then damage in purse after each use. And we were so improved our procedure to define the requirement of rings with. You know, the alkaline solution and the flush with pure water and the pH check before use. So generally speaking, it.

说话人 2 00:17:45

Because of the timeline, we proposed three approach according to our discussion. And each approach, they will be implemented according to the timeline. And then this will impact me, demo batch, pre pvq batch and commercial batch and. According to the increasing requirement, increasing criticality, you know, in a typical batch for the latest steps, for example, in a purification and t tff operation, the approach tool will be used. But in a commercial. In the commercial batch, we will finally switch to approach 3. That's the most preferred methods in the prefer water delivery. So this is a general timeline and plan for the.

说话人 5 00:18:43

Host in the pure water.

说话人 5 00:18:54

Yeah, so maybe just a quick question on approach No. 1. I see there that you. That under the mitigation plan, you have three consecutive days of qualification or maybe what we would call internally validation of those new worst case use points. Is that consistent with like your internal purified water validation procedures in terms of.

说话人 2 00:19:24

Duration. I think I was expecting more like two weeks of sampling for validation. Okay, we understand. It is okay. We understand your point. So from the speaking, this is considered as a tandem qualification because. We have already finished the initial qualification, for example, study 1, study 2, and Study 3. Study 1, study 2, they include both two weeks and study 3 include one year. So for this qualification, that's is a datum qualification for the, you know, for the

multiple host. But the host itself, we have already qualified in the initial qualification in this situation, we only, you know, a poor sample from the worst case of the endpoint. But anyway, we can consider for your suggestion. But anyway, we will define. After, for example, after how many days we can release to use in the product. And finally, we will finish, for example, 2 weeks or one week, maybe 2 weeks. But after 3 days, we can release for use.

说话人 5 00:20:46

In the production. So this is the approach. Okay, that helps clarify it. So you're classifying it as just an addendum as opposed to a new validation. That make sense? Okay, great. That sounds good. And then maybe we're. Only other piece of feedback is, do you guys see approach 2 as being necessary? Or do you think you could just manage approach 1 for a longer duration? I just, it's a very sensitive topic already, and. I guess I see some value in minimizing the amount of iterations that we do. So like to me, it's maybe a little bit cleaner to just do approach one as the interim solution and then jump right to the long term solution without a second interim step. But if you're, if your internal team is signaling that.

说话人 2 00:21:52

Approach one will be unfavorable to manage, then maybe we could agree on also doing approach 2. Okay, we can review the data in their qualification stage in Approach 1. If that's fine, maybe we can skip.

说话人 2 00:22:12

Approach 2. Oh, after approach 1, then we will finally jump to approach 3. That's also a good choice, but we can depend on the data we collected in a qualified stage.

说话人 3 00:22:31

So can, that sounds good. So can I ask, Andrea, like your thoughts on this as well? Are all of these approaches compliant with our like BLA strategy? Cuz like I'm looking at the table at the bottom here and like, as a rule of thumb, I would never like. Choose like different approaches for like a pre PPQ and commercial batch, right? So the notion is that these are timetable driven, right? But like, are they all compliant? Because if they're not, like, if you, eh, like they're, yeah, yeah. Okay. Yeah, they're, it's so like.

说话人 5 00:23:12

0, option 1 is still compliant because we're still supplying water that's been demonstrated to be at quality at point of use. It's just approach one does carry some optics concerns with it because in. Instead of nice, clean, hard pipe networks, you're gonna have, you know, in some cases 25,30

foot of hose slung together while you're processing. So it meets the requirement on paper. It's just more a visual best practice optics thing as to why we wanna move to approach 3 long term.

说话人 2 00:23:53

Okay, now that's all I need there. Thank you, Andrew. Yep. Yeah, thanks for the discussion. But anyway, for BL product and the approach 3 from the speaking list, the most perfect. But anyway, all these three approach, they're compliant, they're offer GMV complaint. The approach 3. The risk is in, you know, less than. Lowest risk. That's why we, you know, finally, we will select Approach 3. That's in a long term.

说话人 5 00:24:32

Okay, appreciate it. And then maybe just the last two questions for me. So on approach 1, just confirming like after you this addendum for the worst case setup, then you'll also move the routine monitoring samples to be from the end of the worst case setup. Correct. Okay, perfect. Great. And then maybe we don't need to discuss it here. I just don't want us to lose it from our radar. And I don't know if there's been separate discussion on the topic, but also in the email I sent out a broader question on when we met with some of our PR and d quality colleagues. The, to get input on this. They had signaled that they were also concerned about other purified water loops for other programs. So again, we don't need to just get discuss it here, but if you could revert back to the email I sent and just look at the bottom of it, there's a question about is this issue only applicable to the G G 9,1,7 process? And other purified water loops that support our other programs are designed, okay, or is this a potentially systemic issue that we need some follow up conversation on for those other programs. Yeah, and it's Dan. Okay, great. Thank you. Okay, thank you.

说话人 1 00:26:04

Right next, let me share this real quick. Alright, is everybody can see my screen. Yeah, yes, can. Perfect. So, yeah, so this is the over overview, like overall status of our pipeline improvements because like this in is include the not only the wild progress and also the passivation progress, it also include the Mo, Moc check, slope and drainage.

说话人 1 00:26:50

For all ops. So as you can see for this, for the in this table, like for example, for the SPS, so for like m MOC and the wild check and slope and drainage, like those are still in progress. And for the passivation, the equipment, the. Equipment, but basically is the main equipment and also the pipeline for the process line is already being passivated and the pipeline for the solvents after

the filter is still in process. So, and for SPPS, the potential risk for this is the some solvent pipeline does not meet the slope and drainage.

说话人 1 00:27:39

Requirements. However, the pipelines are always filled with the non growth promoting solvents. So we identify the risk is low. So. That's why there's no corrective action required. And the due date is for like the all those activities gets completed. So by October 14th for SPPS, and similar approach for. All the other steps. And as you can see, like we have like October, by the October 15th, like all the process steps for the G G No. 7 should be done for all those activities. And one notes for. Or the welding check acceptance criteria we have here is the weld shall be fully welded with complete penetration. No cavitation, pits or cracks are allowed after weld polishing, the internal and external profiles shall be uniform. The transition between weld and base material shall be. Natural, smooth and flat undercuts bury residue while edge distortion, residue oxidation, discoloration and other defects are not permitted. And also for our post passivation work, we have is we do the pH conductivity tests. And also the entire equipment train shall be flushed and with purified water and also sample link for POC and also the IR testing. So any questions on this?

说话人 5 00:29:38

No, I think this looks good, at least from my end. Maybe just a question on the post passivation work there, where you say you're gonna flush down to pull purified water samples for TOC and iron. Maybe just two questions. Do you know off the top of your head or could you share via email what TOC limit or acceptance criteria you're gonna apply? To release the equipment and say that you've flushed it enough? And then secondly, after you get those samples, do you plan to then.

说话人 2 00:30:16

Chase the system with a solvent rinse to displace any water left behind from the water rinse. Yeah, the limit of the QC testing next to be comply with the QC for people for water itself. Perfect, great.

说话人 3 00:30:37

And just to ask them, when you say you're gonna flush the entire system with water, because obviously a lot of the concerns about the passivation involves like a lot of the solvent lines from the bulk tanks. Does that include, do that presumably doesn't include those lines. Okay, good. Okay. Here.

说话人 2 00:31:01

For this table, this is only, you know, summary of the pipeline after the filtration. After the filtration, inner, how to say, inner s. So in the solvent pipeline, okay, that's include, you know, the solvent pipeline after the as filtration. And the pipeline for the solu, for the process solution between different equipment. And are there sections of the pipework prior to that filter that are being passivated for the filter, for the PA, for the passivation of the, so the pipeline before the filter, that's how to say, according to our discussion, first important, that's the. Header. Of the tank. We checked the header one by one and documented inner checklist. And then we screened each of the how to say, the, the welding location. To identify if that's, if any additional work is needed for the preservation of the entire software delivery pipeline, you know, that's how to say.

说话人 3 00:32:47

That will be GI did additional time because you know the system, that's a big system. And yes, I understand. I just wanna make sure, because I know Asim Kim does a level of iron monitoring. Yes. And it's bulk tanks, right? Independent of the work that we're talking about here, right. But I just wanna make sure that between those studies and the studies we're talking about here, that we have the entire. System mapped for potential iron leaching into water, if you understand my meaning, so that it's all, so there's no gap. So I understand this bit is looking at from the filter onwards, right? But I just wanna make sure.

说话人 2 00:33:47

That there's another body of work that covers everything from before the filter, if you understand. Yeah, I understand for the pipeline before the filter, as we discussed in the previous meeting, first, never the routine monitoring of the software in the tank.

说话人 2 00:34:10

For the iron, this is the first assurance. Another assurance is, you know, the pipeline, that's the same pipeline what we used for the, you know, for the, for the, the, for the first batch. And we screened the pipeline intended for the preview batch and the first demo batch. They're the s, exactly the same pipeline. So this provided additional assurance of the acceptance, credit acceptance of the pipeline before the filter. Okay, thank you. Okay, you're welcome.

说话人 6 00:35:03

And I have a question here to see, because I know that Andrew ask about it's like, I know that the both passivation work is so you're going to the passivation that is going to be put like flush with water, there is going to be a pH test and conductivity to see if everything is good. And also, Aaron, but after that, from my other student, and you can correct me, there will be like a solvent clean to ensure that the water. That was use is completely release. Is there going to be a test?

说话人 2 00:35:30

For water content to verify that there is no water there. Okay, in this way, AC, according to current practice, after the first, the answer is yes, because at the end of the, you know, the posted participation work. After the flash, we have to, for example, rinse with solvent and then dry. Yes, for use. If we don't do this work, there will be some red to WA, potential red to water in the pipeline. This will bring, you know, some side effect to the, you know, to the mixture. Okay, good.

说话人 5 00:36:23

Alright, do we have any further questions on this topic? Alright, if not, let me. Hey, Alex. Yeah, sorry, no remaining question on that topic, but I just thought of one additional question for the purified water topic before we go on to the next one. So while we're working on Approach 3 and retrofitting those hard pipes, headers that exist, will there be any sort of segregation of those headers from the tanks they feed? Just like to ensure that they cannot be used until Approach 3 is release. So like for example. Disconnecting those lines and blanking them at the tanks, instead of leaving them connected in the meantime, if that question makes sense.

说话人 2 00:37:20

Yeah, we understand your point. I will say yes, the, oh, the current collection, they will be discontinued. And then we, that's why I said we have to improve our procedure to allow the preferred.

说话人 2 00:37:43

Strategy for the, either the approach 1 and approach 3 to allow the drain flush and cleaning only after work. Those are can are completed. We can proceed with the, how to say, the approach 3. Okay, sounds good. Thank you. Okay, you're welcome.

说话人 1 00:38:18

Right, so the next topic I gonna share is basically the global call use and removal timeline. So as you can see, this is the list of all our steps and also equipment and. Basically, from this column, you can see like if the equipment is new equipment use for the G G 9,7, or it has been used for use for other projects before. And also, and as you can see, like, and when. The glycol was used during the PQ stage, and also the concentration of the glycol solution and also the last cleaning time of the equipment.

说话人 1 00:39:17

And for control measure will do level 2 and TOC and. And this so will be the complete dates for for for that for the L2, level 2 and also the POC.

说话人 1 00:39:40

So this topic, this is a follow up topic according to our.

说话人 2 00:39:45

Previous discussion, in a previous discussion, we talk about whether the Galico are used in their equipment qualification. So in a discussion, we said we have to check afterwards. And if let's use, we have to, we will test the QC. So after our offline screening in a computer cleaning and the Galico, indeed, it's used in a equipment qualification. So here you can see in the last two column, we planned for the TOC testing before release to use in this G G 9,1,7.

说话人 3 00:40:39

Any questions on this topic? So, sorry, let me understand this here, right? So this is the entire equipment set. Yep. And you're telling us that during the qualification, there was ethanol, glycol being used, these various different concentrations, and these are the last, these are the subsequent recorded cleaning times or the last reported cleaning times. For these various different bits and pieces, and then you're saying that each of these were applicable with some exceptions, was subject to testing via the TOC, and presumably those all passed?

说话人 2 00:41:29

Is that we will do the QC testing and a only after it's passed, we will release to use in G G 117. Okay. And but the there's obviously there's no results on this table is just reporting about when the cleaning.

说话人 3 00:41:52

Occurred and when the testing was. Sorry, the testing is to be done. Sorry, this is the upcoming testing, right? Was there ethanol, glycol, was there TOC testing done on the step 1 equipment prior to the demo batch? That was executed on that equipment train. You don't have to answer that now. You can go in and figure that out. Looks like some, that was. But not, maybe not all of it. So if you can go away and answer that question, that would be good just for the other investigation. But yeah, I'm absolutely in favor of you conducting additional testing on the equipment train prior to further use on it for confirmation that there's no athlete like call present there. But it would just be good to know.

说话人 2 00:42:47

For the work that we've done already. If the equipment train is, sorry, if the equipment train was cleared for use and had no ethanol glycol in it prior to using the demo batches. I can see this way for the completed batch, you know, because the equipment is not a new equipment after qualification, there's other product that they have already run. In equipment, so that's not in there. First product after equipment qualification. So for that information, so sorry, there was other products run in this equipment that after qualification. Okay, wait a moment. Let me confirm.

说话人 2 00:43:40

在 demo batch 之前，一线咱们已经完成了批次之前。设备在验证完毕之后呢，设备有其他项目生产过，才用到他项目上嘛。已经完成的。已经完成的大多都是新设备，除了那两个以外，那两个是用过其他项目。不是的，已经完成了，这是将要用的，已经完成那个批次。只有一个设备用过其他项目，其他都是新的。新设备是吧？对。只有一个设备是。

说话人 2 00:44:13

Okay, for the completed batch, you know, majority of the equipment there are fresh equipment after qualification, we cleaned and used in your product, but never the one I, one piece of equipment. After qualification, let's usually in 0 other product, and then usually in your product.

说话人 3 00:44:41

what was that piece of equipment?

说话人 2 00:44:44

这是个哪个设备呢？一。H 列，H 列那个还没用呢，那第三步呢？不是说已经完成的。

说话人 1 00:44:54

已经完成的。已经完成的。

说话人 2 00:44:57

对，之前做了，这次又起来了。

说话人 1 00:45:01

不 scale 的批次。

说话人 2 00:45:02

已经完成的批次。

说话人 4 00:45:03

啊，上面不是说，你给我写的，给销管使用前是设备状。

说话人 2 00:45:19

Okay, for the finished batch, they can summary into one table similar with this one. So after that, we can provide to you that that will be much clear. Okay, no worries. Thank you. Yeah, but anyway, after equipment qualification with glyco, we cleaned with pure for the water and then released for use. But here, as we planned, additionally, we will do the QC testing.

说话人 6 00:46:03

Before we needs to use in TT eleven 7. I have a question regarding that QC testing because I see the control measurement is TOC. Are you going to. To be conducting also a TAMC and tymc test to verify micro growth and also endo test before releasing the equipment.

说话人 6 00:46:27

Let me know if it makes sense. Okay, because I'm seeing the control measurements just to see on L2. I don't know what exactly L2 means, but I just want to verify, okay, L2, that means you, you know, the quantitative testing of the previous product if there is. Okay, perfect. And then L2, how it's going to be exactly conducted will be a BCA test or what is going to.

说话人 6 00:46:51

B. That's the, that can be done by QC or by Spc. Okay, okay, okay, got it. I'm regarding micro test is to release the equipment. Are you going to be conducting our micros test to ensure that there's no any micro growth or anti toxing leftover.

说话人 2 00:47:15

ACO, I will say this is according to the Clean Redden protocol. Typically, we will do to the from the, how to say, from the clean operation for. Be from the purification step. Okay, so it's this like purification step. From my understanding, there will be a micro test, right? Correct. Okay, okay, okay, got it. Because you know, you know, previous early steps, you know, that's risk is not high.

说话人 6 00:47:52

We will do the, you know, the microbell testing of equipment starting from the perfect step. Got it. And it's like that micro test is good. It will be used to release the equipment before even the BQ BQ batch begins or demo batch, right? Correct. Okay. Okay. Got.

说话人 6 00:48:20

Right. Any further questions on this topic? I have their question. Sorry, every guarding that micro test, everything like POC, all the parameters or all the specs parameters of what are going to be the ranges of what you're expecting. Those are going to be included in the in the document, the cleaning document, right?

说话人 1 00:48:40

Yeah, that would be included in a clinic. Okay, perfect. Document the test, outcall. Perfect. Good.

说话人 1 00:48:52

Alright, thank you. Think I do have another topic I want to discuss with Lily's team. Let me share my screen again. Alright, so I think probably this question is more for Andrew, so basically we know like we talk about this like about the filter integrity test that FIP. So like, I guess go over this for real quick for current process like we for pre FIP that's being done by the suppliers and. We like ACAM does not conduct pre fit in house. And for the post fit, we do that after the everything is done. So well, we know like during the outside visits, you talk about like, since we buy the cartridges is not the like. We install the cartridges to the filter housing, so there will be like a potential like risks, but basically, like for example, improper installing of the oring may fail the fit. And so right now. I think one thing we want to discuss with Lisa team is like for the post integrity test, we have no problem. However, for the pre integrity test, we do have some challenges because like the primary concern is the integrity test typically introduce like non process solvents such as like IPA or water. So like, like poor removal of this solvent may lead to cross contamination from introduced foreign solvents. So therefore, like removal of those test solvent is very critical. So. There's a like, or team came up, like there's two ways to remove that. The one way is to, is the driving method. So like drive the entire filter assembly. And according to the com, according to communication with the supplier, the supplier only drives filter cartridges and exclude the filter housing. So this approach maybe will full address Lily's concerns because we'll, if we can drive the whole filter, but study is required for the full filter assembly driving method. And also relevant driving equipment is not currently available. So cleaning area, clean area modification will be necessary with estimate lead time of. Three months, which requires like timeline assessment. That's the drawing method. Another method is like solvent displacement method. So basically use process solvent to displace the test solvent. So the as effectiveness of the displacement requires validation. And the intensive laboratory

testing is needed to establish acceptance criteria for residue test solvent. Since these filters are installed at multiple locations, this will kind of like increase like a great amount of the testing workload. So right now, like our team still propose to rely on the supplier COA for pre use filter integrity test. And we will definitely implement like. Do personal verification to make sure the filter cartridges installation is correct and perform like integrity test post use. So just want to check with Lily's team like see like if that's okay or like what's. Will be like best practice from your end to kind of like solve this problem.

说话人 5 00:53:26

Yeah, I think I don't have any immediate concerns. I think maybe just a few points. The first being that we still are required to do like a post installation flush to kind of serve two purposes. 1. Sort of functionally close that filter installation just cuz it was a briefly open operation, but then to serve as an extractable leachable removal flush for the new element. So just confirming that even and if you're doing a pre use test, the 70% IPA that you use for the pre use test basically serves that dual function. Like when you wet the membrane anyway, you get that benefit. So just confirming that even if we move away from the pre use test, that you'll still plan to do maybe a methanol flush through the newly installed elements directly to waste before use. Okay.

说话人 1 00:54:31

Okay, I'll check with.

说话人 5 00:54:33

My team see like if that's something like we can do. Okay, so that be my first question. And then my second question would be like, I think there is a little bit of risk here, right? I agree it's low, especially if you incorporate the SPV. But there is a little bit of risk that you get a failing post use because of an installation error that you didn't detect. So I think we would just need to understand like, are we okay accepting that risk moving forward and the implications it could have on.

说话人 5 00:55:14

Batch disposition, or do we wanna further buy down risk? I think a way that I've seen people further buy down risk when they're in this situation is they'll actually have two elements in series to one another, so that only one of them needs to pass the post use and you would basically need two redundant failures. During install to actually fail. So I don't know if that is out of the question or if that's something that. Your team could look into feasibility of adding a second filter, basically back to back with the first to further bed down the risk. Or maybe we can

all come together as a team and say that in the event we had a feeling post use, we feel strongly enough about being able to overcome that result with technical justification to not.

说话人 5 00:56:11

Toss a batch. Cuz I think the problem too is like the filter, the other option is. You leave just one filter, but then you leave yourself a method to transfer backwards in the process. So let's say you failed your integrity test, you could transfer back from your dropout tank back to, say your retentate tank and then new filter in and then re execute the filtration.

说话人 5 00:56:44

I think the problem historically with that, it depends a bit on the submission language and the registration language, but I think that's considered reprocessing. So then that comes with its own nightmares. So it's best to just avoid that. And try to buy down risk to an acceptable level. So I think those would be my only thoughts is how are you gonna handle the functional closer closure slash Enel flush? And then do we think risk as a team, do we think risk is low enough with the SPV of the filter installation to buy down the risk of a post use failure.

说话人 2 00:57:27

Or do we want to look at adding a second filter in series to buy down that risk? Okay, here we understand your concern. You know, from the speaking, if a redundant filter is use, there will be, you know, to reduce the risk. And this is first idea. So another idea is we do the pre use integrity test. But here, considering, you know, there is some or. Practical difficulties to s, to replace the solvent, which are used in their integrity test. So did you have some experience on how to. Replace the solvent, which are used in the integrity test before the.

说话人 5 00:58:44

Filter is used in the process. Yeah, I agree with you that I agree with your sentiments on the slide that in this case, it would be a bit challenging and you would have to essentially, yeah, validate a removal method. And it's not quite as straightforward like we're used to. We typically use hydrophilic membranes at, for these final filtrations. So then you're just testing in water. So then when you do a solvent flush to displace the water, we use simple things like. Density or Ri to prove removal.

说话人 5 00:59:30

Same thing with like the TFF membranes when you're flushing the ethanol out of them that they come stored in from the vendor, you're flushing it out with water. So same thing, you can use like a simple Ri or density demonstrate removal. Here it's a bit trickier because you're using a

hydrophobic membrane. That you're gonna have IPA, water, and methanol. So you're gonna have a three component things. So I agree it's gonna be more challenging and there's not gonna be a quick and easy way to demonstrate removal.

说话人 5 01:00:01

Cuz that's typically what we do is we would just use like a density or RI check to demonstrate removal. But that wouldn't work here because you wouldn't be able to readily discern between residual IPA and the methanol. So I, I agree. I understand your concerns with why it's challenging. But honestly, that's another reason that we also prefer the capsule style. Filters instead of the cartridge where it comes as one unit. Cuz then you don't have to do the pre use test. So I think I'm personally okay with what you're recommending here as long as everyone on the team can feel good about it. If it's possible for you to add a redundant element, I think that just further strengthens the control. And buys the risk down to next to zero, if you can, then I think we just need to understand that risk that is there and make sure we at least know how we would respond to a failed post use test. Yeah, we understand that maybe according to your comment, maybe in the best way is.

说话人 2 01:01:15

The best way or the practical way is we use a, a, you know, a capsule kind of, capsule type of filter and then followed by a redundant filter that will be, you know, mitigate the risk.

说话人 5 01:01:37

Yep, yep, I agree. Yeah, so if you can look into those as options, that's great. And then, but if they're not possible either, I think what you've put some really good thought into this. And I think with the SPV control around the installation, etc. I think, yeah, I think we've got several good options on the table. So I appreciate you guys looking into this.

说话人 1 01:01:57

Okay, yep, Andrew, quick question. So, yeah, like we also, our manufacturing team also look into the one piece, the capsule sell like filters, but like, you know, like, you know. The types we look into, they're not like explosion proved. So that's why we can't like hard to find like the kind of like non static filters. For like one piece filter for in your case. So do you have any experience to have like those type of like one piece filter? Like there's like no introduce static like in in the workshop. Yeah, let me send you I can send you some manufacturer and part number combos that hum.

说话人 5 01:02:57

We have history with. And just so you can see kind of what we typically use and see if it if you can find like an equivalent. Yeah, so I can okay, email those over. Okay, perfect. Yeah, I'll will definitely look into that because I think we want to, you know, like try to use the one piece, but.

说话人 1 01:03:17

The challenge right now is like we cannot find the right type to serve, you know, like, or workshop in at the moment. Okay, yep, that sounds good. I will send over. Okay. Okay. Thank you. Thank you. This sounds good. Yep, usually. Alright, and Alex, sorry, I had one more question. It says that the post use.

说话人 7 01:03:39

Is testing is done after disassembly? Is that the disassembly between batch to batch cleanings? Or what is the frequency of the disassembly? Can you put me with you with my opinion? Automobile condition, Tina. So depends on like different filters. We have like different frequencies. Okay, so I if we have a post use filter test that fails sometimes that means multiple batches will be implicated and then sometimes it would be just a single batch. Is that correct?

说话人 2 01:04:20

Oh, that should not correct. That should not be the case. The impacted batch between, you know, the di, between the, the you, between the initial installation and the last time integrity test, they're all impacted batch.

说话人 5 01:04:40

They will be evaluated of as a risk assessment. Yeah, so maybe I'm not quite understanding and maybe we haven't talked through this detail, but I think at least our historic expectation would be that it's a 1 to 1 between filter and API batch. So I have seen it allowed that like in our case, for example, we have multiple TFF sublots. That are gonna make up a single. Afd sub lot, then that's okay to run multiple TFF sub lots, but it should be one filter to one API batch. And the control for that usually is we typically predicate precipitation start on a passing post use test. So you filter the material between the TFF for 10 take tank. And the t F F dropout or product hold tank. And then you don't let precipitation start until you have a passing post use test. Because if you let precipitation start and you make a slurry, you have no recourse where if you have it as dropout material still.

说话人 2 01:05:54

And you get a failing result that you can't overcome, you do have some options for how to handle the material before you precipitated out. Yeah, understand only the integrity test that

passed, we can proceed with their participation operation. Yeah, I have a question here because you mention some, the key thing, Andrew.

说话人 6 01:06:17

I was very fine, like the diagrams and what I have been seeing is like in the case of the TFF, like after you finish the retained tank, it will go to the other tank. That is this, I don't remember the specific name, is salt. And I know that in this, in that South Tank, south formation, I think is the name, there's an AOH is going to an AOH with methanol is going to be added to the system. And then the final filter for point 22 microns is before the preset tank, not between. Yeah, that was, yeah, that was, yeah, that's like him.

说话人 5 01:06:52

But you mention it's like a risk because it'll be. Yeah, the yeah, the, I don't know, the diagrams have been updated. We caught that several weeks ago and that filter has been moved from, it's been moved to between the retentate and the dropout tank as opposed to after the dropout, before the precept tank.

说话人 6 01:07:13

Got it. And the ACM Cam team can confirm me on that. But yeah, we discuss that a few weeks back. Got it. But to be honest, that brings me a concern because I know that in the drop is like when the caustic methanol solution is going to be added, it's going to be in the last tank right before the preset.

说话人 5 01:07:33

Does that present a risk? No. It's just like we would have the same con. So like, I'll, yeah, so like, think about like you add like your cake wash solution after the final filter, you add your anti solvent at pre sip after the final filter. So you have solutions that you add after the final filter. They'll just have their own filtration controls on them. Okay, got it. Yep. So it'd be just like that. And Naoh and Methanol's non growth promoting low risk solution. So yeah, yeah, perfect.

说话人 5 01:08:12

So a few follow UPS on this topic, but I think we're really close on it. Okay. Yeah, so to recap mostly for myself here, it was making sure that if even if we're not doing a pre use test that we're gonna do like a methanol flush for functional closure. And E&L removal to waste before use, you'll investigate the opportunity to add a second element in series to help further buy down risk. I'll send along some of the capsule style elements that we have internal. Experience with in this sort of operating environment for you to review, and then you guys will just double confirm

that the filter has the final point two has been moved to between the TFF system and the product hold vessel and it's no longer between, it's no longer on its way to precipitation. It's. It's before the product hold vessel that feeds precipitation and our expectation, okay, and then our expectation there again is that you don't start precipitation until you have a passing post use integrity test. And that requirement drives a 1 to 1 relationship between.

说话人 8 01:09:35

Filter element and API batch. Yep, gotcha. Okay, sounds good. Thank you. Yep, thanks. And then I know we're a fair bit over time here, so one final question. Given all the dates and these activities, you know, outside of the standard deliverables. What date is the Asim Kim team now targeting for purification demo start? Is that the week of October 19th or the week of October 26th? Just again for planning purposes here as we march deliverables along.

说话人 9 01:10:16

give me one second, 喜哥喜哥在吗?

说话人 2 01:10:21

在找。

说话人 9 01:10:23

他 first to demo, 我们现在是他给什么时候?

说话人 2 01:10:27

他给的是应该是20号。

说话人 9 01:10:30

Yeah, October 20th. Hum. Okay. Yeah, it would be the day after the indicated date for the water system remediation there for approach 1. Okay. Okay. And just wondering if you guys still have more time to extend this meeting a little bit or. We can just end it here. Like I can send you like a update for the some of the by preparation Powerpoint. Yeah, I could potentially stick around. I'm not sure about others on the line here or if we just wanna take it offline.

说话人 5 01:11:13

Yeah, I don't know about Steve for if he's still on the call. I do have to draw my head and I had a 9 a.M. But I can definitely review slides offline and yeah, sure. And I did have a, and yeah, and one

quick question on those slides, cuz hopefully they have some more of the reverse phase fractionation information.

说话人 5 01:11:34

One question I have, did have is if you could provide the path length for the UV detector that you use in the lab to generate the data. Cuz I think you give the Millie AU cut in point. I just wanna convert that to like a normalized value, like an AU per centimeter. So I was just wondering if you could provide the path length for the UV flow cell used in the lab. Yeah, sure. I will later send you an email with the pass language. Okay, great. That sounds good.

说话人 5 01:12:07

And then I think the only other pressing thing I had from like a process technical discussion before I have off here is I think we were still waiting on you guys to send the agitator information and vessel geometry information for the precipitation tank, just so that we can. Confirm power to volume is met for the precipitation reaction. So I think you were gonna send us, you know, like the dimensions of the precip tank, but then also size and shape type of the impeller used. Yeah, let me double check with my team to see if we can get you something these two days.

说话人 8 01:12:45

Okay, great. Thank you. Hum. Okay. Yes, we could provide those offline. I know we've lost a number of folks from Lily team here since we've gone over, but we can look, follow up via email or similar here. So, okay, thank you and you guys have a nice day. Thanks everyone. Thank you.